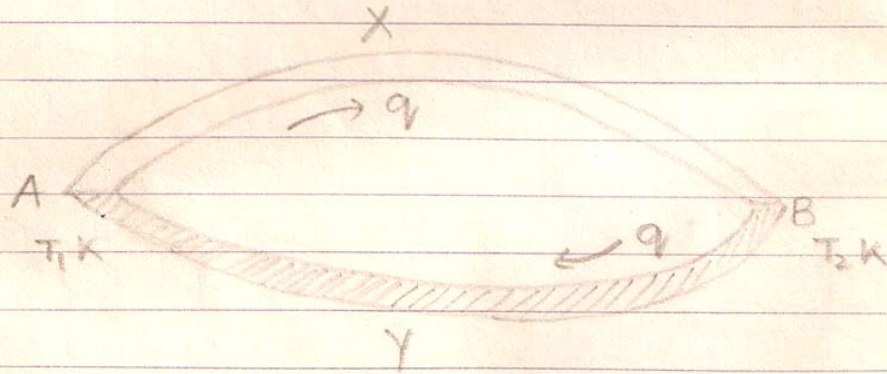




Thermodynamics applied to thermoelectric effect:
Thomson effect:



Let us consider a thermocouple consisting of elements X & Y and the temp. at the two junctions A & B be $T_1^\circ\text{K}$ & $T_2^\circ\text{K}$; ($T_2^\circ\text{K} > T_1^\circ\text{K}$).

Let us consider an elementary positive charge $+q$ be taken through one complete round along the circuit, at an infinitely low speed ($i = \frac{dq}{dt}$) corresponding to a vanishingly small current^{dt}; so that irreversible Joules heating effect ($H = i^2 R t$) can be neglected.

Thomson Effect



Let for the given thermocouple:

π_1 & π_2 be the peltier coefficients at the junctions A and B respectively.

As the charge 'q' crosses the junctions, work is done

(i) Work done on the charge 'q' i.e. heat energy absorbed as the charge crosses the junction A = $\pi_1 q$.

(ii) Work done by the charge 'q' i.e. the heat energy evolved as the charge crosses the junction B = $\pi_2 q$.

The net work done by the charge 'q' as it makes one complete round the circuit = $\pi_2 q - \pi_1 q$.

Work done per unit charge = $(\pi_2 - \pi_1)$ = Seebeck e.m.f. developed in the circuit = E.

$$E = (\pi_2 - \pi_1) \quad \text{--- (1)}$$

Using the definition of entropy:

(i) Change in entropy at the junction A = $ds_1 = + \frac{\pi_1 q}{T_1}$

(ii) Change in entropy at the junction B, $ds_2 = - \frac{\pi_2 q}{T_2}$

Since the cyclic process (of the charge moving round the circuit) is a reversible process; applying the law of thermodynamics, the total change in entropy must be zero.

$$\text{i.e. } ds_1 + ds_2 = 0 \quad \text{or} \quad \frac{\pi_1 q}{T_1} - \frac{\pi_2 q}{T_2} = 0$$

$$\frac{\pi_1}{T_1} = \frac{\pi_2}{T_2} \quad \text{or} \quad \frac{\pi_2}{\pi_1} = \frac{T_2}{T_1}$$

Thomson Effect



$$\text{or } \frac{\bar{\lambda}_2}{\bar{\lambda}_1} - 1 = \frac{T_2}{T_1} - 1$$

$$\text{or } \frac{\bar{\lambda}_2 - \bar{\lambda}_1}{\bar{\lambda}_1} = \frac{T_2 - T_1}{T_1}$$

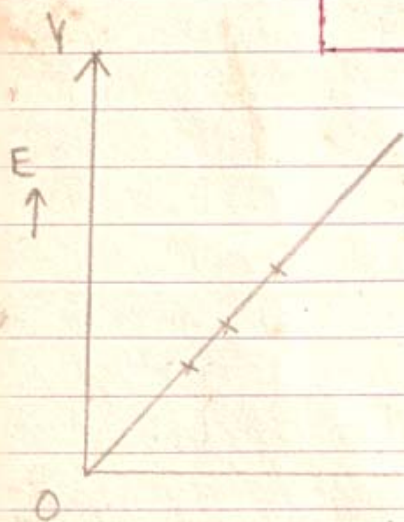
$$\text{or } \bar{\lambda}_2 - \bar{\lambda}_1 = \frac{\bar{\lambda}_1}{T_1} (T_2 - T_1) \text{ --- (2)}$$

$$\text{putting eqn. (2) in (1): } E = \frac{\bar{\lambda}_1}{T_1} (T_2 - T_1) \text{ --- (3)}$$

If the temp. T_1 of the cold junction is kept constant $\bar{\lambda}_1$ also remains constant. On changing the temp. of the hot junction T_2 ; $(T_2 - T_1)$ changes & E changes.

from (3):

$$E \propto (T_2 - T_1)$$



Thus according to eqn (3) the variation of Seebeck e.m.f. with temp. difference is a straight line, but the experimental results show a parabolic variation.

This discrepancy between the theoretical result and the experimental result was explained by Lord Kelvin; who suggested that some new change must also take place in the process → which led Thomson to discover Thomson effect.

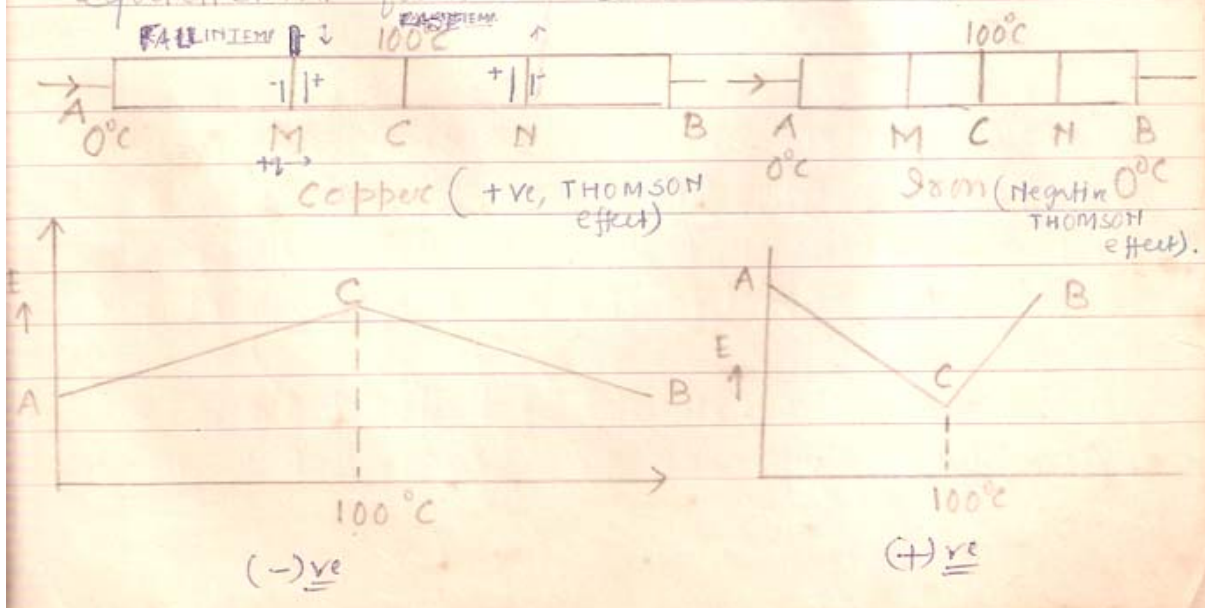


Thomson Effect

Thomson effect: When current is passed through an unequally heated conductor, heat is absorbed or evolved at different points along the conductor. This phenomenon is known as Thomson effect.

Explanation: In an unequally heated copper bar, point of higher temp. corresponds to lower electron density & hence higher potential. Hence two neighbouring points being at different temp. will be at different potentials & so a small p.d. is set up between the neighbouring points. On passing current; as the charge flows through that p.d. work is done by the charge or on the charge i.e. heat is evolved or absorbed.

Demonstration: One copper bar (& one iron bar) of uniform cross-section is taken, with its two ends immersed in melting ice i.e. Temp. at the two junctions is 0°C . The mid point 'C' is kept at constant temp. 100°C . M & N are two points equidistant from C. Two holes are drilled at M & N.

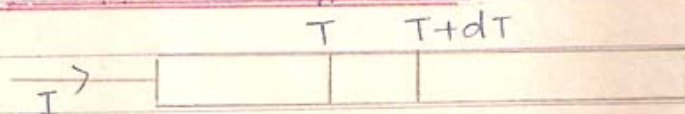


Thomson Effect



and two thermometers are inserted into those hole; which show same temp. (say 80°C).
 Current is passed through the bar from A to B.
 In copper bar from A to C, +ve charge flows against the P.d work is to be done on the charge i.e. heat energy is absorbed; temp. at M falls. From C to B the +ve charge flows in favour of the P.d work is done by the charge; heat energy is evolved, temp. at N raises. The thermometers inserted at M & N also show fall in temp. at M & raise in temp. at N.
 In case of Iron just reverse changes can be observed i.e. on passing current in the same direction; temp. at M raises & temp. at N falls.
 Thomson effect is said to be +ve in Copper & negative in ~~Iron~~ ^{Iron} & is probably zero for lead.
 Ag, Zn, Cd, Sb also shows +ve Thomson effect.
 Pt, Ni, Co show negative Thomson effect.

Thomson coefficient :-



Let us consider two neighbouring points at temp. 'T' and 'T+dT'

Let σ be the P.d per unit temp. difference.

$\therefore \sigma dT =$ P.d between the two neighbouring points.

Let a current I be passed for t sec.

Work done as the charge flows between the two

Thomson Effect



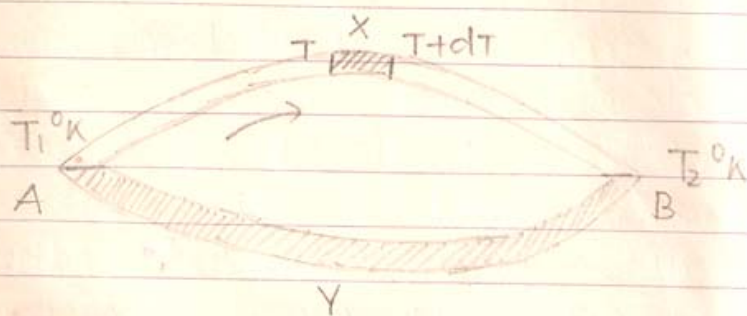
points is heat energy absorbed or evolved between the two neighbouring points. = $(It) \sigma dT$.

σ is known as Thomson Coefficient.

Hence Thomson Coefficient can be defined as work done per unit charge per unit temp.

difference is the heat energy absorbed or evolved when unit +ve charge flows between two points having unit temp. difference. unit = volt/ $^{\circ}\text{K}$.

Thermodynamics applied to Thermoelectricity :



Let us consider a thermocouple X and Y. The temp. at the junctions A and B are respectively $T_1^{\circ}\text{K}$ & $T_2^{\circ}\text{K}$ ($T_2 > T_1$)

Given:

π_1 and π_2 = Peltier coefficients for the given thermocouple at junctions A and B respectively.

σ_x and σ_y = Thomson coefficient of the elements X and Y respectively.

Let us consider a +ve charge q be taken through one complete round the circuit very very slowly, corresponding to a



Vanishingly small current. So that irreversible Joule heating effect can be neglected. Let us consider the motion of the charge from the point A in the element.

The work done on the charge i.e. heat energy absorbed when the unit +ve charge flows from point A to B, through the element X.

$$= \int_{T_1}^{T_2} \sigma_x dT \times 1 \quad \text{--- (1)}$$

Reaching the point B; the charge crosses the junction from element X to Y.

The work done on the charge i.e. heat energy absorbed; when the unit +ve charge crosses the junction $= \pi_2 \times 1$ --- (2)

The charge then flows from B to A through the element Y and the work done by the charge i.e. heat energy evolved :-

$$\int_{T_2}^{T_1} \sigma_y (1 dT) = - \int_{T_1}^{T_2} \sigma_y dT \quad \text{--- (3)}$$

Reaching the point A; the unit positive charge then crosses the junction and goes from element Y to X to complete the cycle.

The work done by the charge i.e. heat energy evolved; as the unit positive charge crosses the junction $= -\pi_1 \times 1$ --- (4)



Total work done by the charge is heat energy evolved; so the unit +ve charge crosses the junction.

Total work done so the unit positive charge makes one complete round along the circuit.

$$= \int_{T_1}^{T_2} \sigma_x dT + \pi_2 - \int_{T_1}^{T_2} \sigma_y dT - \pi_1 = \text{is}$$

Known as Seebeck e.m.f in the circuit.

$$\therefore E = \pi_2 - \pi_1 + \int_{T_1}^{T_2} (\sigma_x - \sigma_y) dT \quad \text{--- (5)}$$

The change in Entropy; during the above four steps are

$$(i) ds_1 = \int_{T_1}^{T_2} \frac{\sigma_x dT}{T} \quad (ii) ds_2 = \frac{\pi_2}{T_2} \quad (iii) \int_{T_1}^{T_2} \frac{-\sigma_y dT}{T}$$

$$(iv) - \frac{\pi_1}{T_1}$$

Since the flow of charge through one complete round along the circuit is reversible according to law of Thermodynamics the total change in Entropy is zero.

$$\therefore ds = ds_1 + ds_2 + ds_3 + ds_4 = 0$$

$$\text{or } \frac{\pi_2}{T_2} - \frac{\pi_1}{T_1} + \int_{T_1}^{T_2} \frac{(\sigma_x - \sigma_y)}{T} dT = 0 \quad \text{--- (6)}$$



Thomson Effect

Let the temp T_1 of the junction A be kept constant. T_1 also remains constant. The temp T_2 of the junction B is changed; temp difference changes and as a result Seebeck e.m.f changes.

Differentiating Eqⁿ (5) w.r to T_2 :

$$\frac{dE}{dT_2} = \frac{d\bar{\pi}_2}{dT_2} - 0 + \frac{d}{dT_2} \int_{T_1}^{T_2} (\sigma_x - \sigma_y) dT$$

$$\frac{dE}{dT_2} = \frac{d\bar{\pi}_2}{dT_2} + (\sigma_x - \sigma_y) \quad \text{--- (7)}$$

Differentiating Eqⁿ (6) w.r to T_2 :

$$\frac{1}{T_2} \cdot \frac{d\bar{\pi}_2}{dT_2} + \bar{\pi}_2 \left(-\frac{1}{T_2^2} \right) - 0 + \frac{\sigma_x - \sigma_y}{T_2} = 0$$

$$\frac{1}{T_2} \left[\frac{d\bar{\pi}_2}{dT_2} - \frac{\bar{\pi}_2}{T_2} + (\sigma_x - \sigma_y) \right] = 0$$

$$\text{or } \frac{d\bar{\pi}_2}{dT_2} - \frac{\bar{\pi}_2}{T_2} + (\sigma_x - \sigma_y) = 0 \quad \text{--- (8)}$$

Putting eqⁿ (7) in (8) :-

$$\frac{\bar{\pi}_2}{T_2} = \frac{dE}{dT_2}$$

$$\text{or } \bar{\pi}_2 = T_2 \times \frac{dE}{dT_2} = T_2 \times \left(\frac{dE}{dT} \right)_{T_2} \quad \text{--- (9)}$$

Where $\frac{dE}{dT} = \left(\frac{dW}{dt} = \text{Power} \right) = \text{Rate of}$

change of Seebeck e.m.f with temp. is known as thermoelectric power.



∴ $\frac{dE}{dT_2} = \left(\frac{dE}{dT} \right)_{T_2}$ = the thermoelectric power at temp. T_2 .

Using eqⁿ (9) Peltier coefficient of a junction can be calculated.

$$\pi_1 = T_1 \times \left(\frac{dE}{dT} \right)_{T_1} \quad \& \quad \pi_2 = T_2 \times \left(\frac{dE}{dT} \right)_{T_2}$$

Differentiating eqⁿ (9) w.r to T_2

$$\frac{d\pi_2}{dT_2} = \frac{dE}{dT_2} + T_2 \frac{d^2E}{dT_2^2} \quad \text{--- (10)}$$

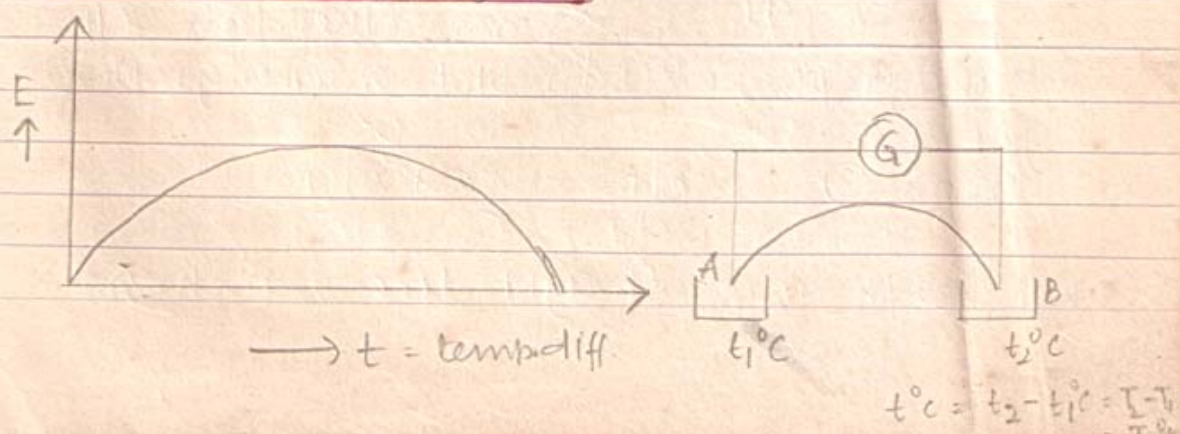
Putting eqⁿ (7) in (10) :-

$$\frac{dE}{dT_2} - (\sigma_x - \sigma_y) = \frac{dE}{dT_2} + T_2 \frac{d^2E}{dT_2^2}$$

$$\text{or } \boxed{(\sigma_x - \sigma_y) = -T_2 \frac{d^2E}{dT_2^2}} \quad \text{--- (11)}$$

Using eqⁿ (11) Thomson's Coefficients can be calculated.

Thermoelectric diagram:

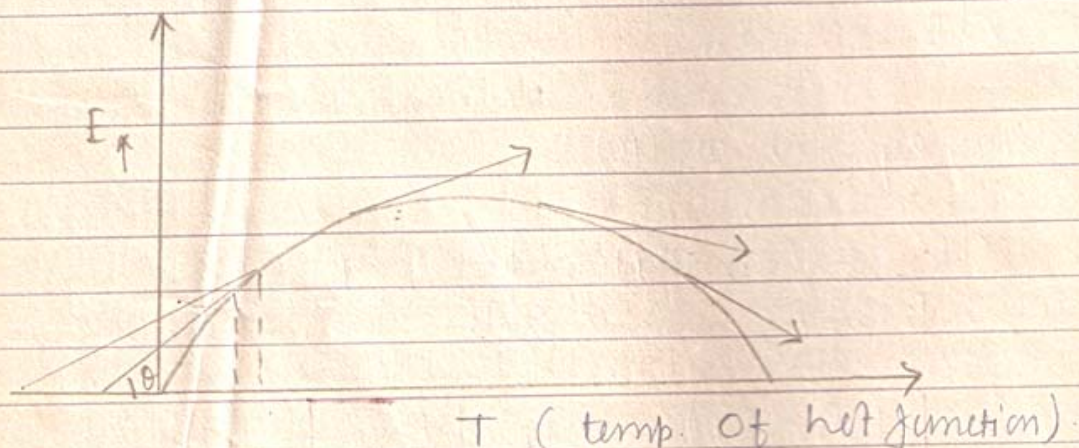




Let $t^{\circ}\text{C}$ be the temp. difference between the two junctions. We know experimentally that variation of Seebeck e.m.f with temp. difference is parabolic in nature and hence can be represented by a quadratic equation.

$$E = at + bt^2 \quad \text{--- (12) where } a \text{ \& } b \text{ are two unknown constants.}$$

$$\frac{dE}{dt} = \frac{dE}{dT} = a + 2bt = a + 2b(T) \quad \text{--- (13)}$$



From $E-T$ graph we can draw tangents at some chosen points & the slope of the tangents at those points (temp) gives the value of $\frac{dE}{dT}$ corresponding to those temp.

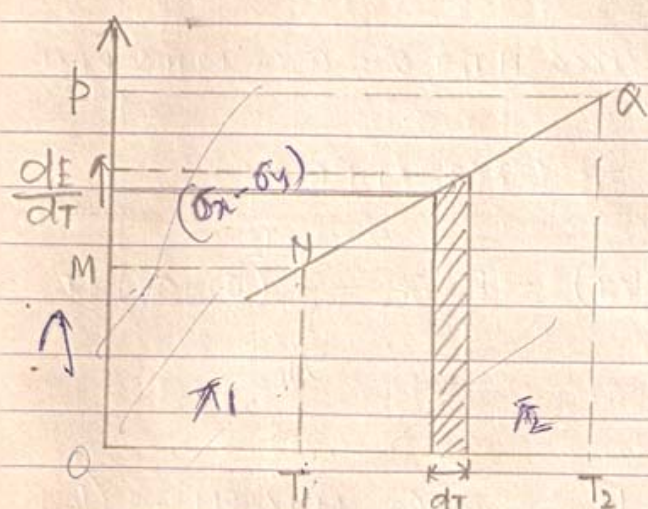
From the data collected from the graph, we can plot another graph, $\frac{dE}{dT}$ against T .

From (3)
$$\frac{dE}{dT} = a + 2bT$$

The graph is a straight line & is known as



Power line. The intercept of the power line on Y-axis gives the unknown constant 'a' and the slope of the power line gives '2b' & hence 'b'.



Let us consider two neighbouring points in the power line at temp. T & $T+dT$. The points are so close that $(\frac{dE}{dT})$ at both the points can be assumed to be same.

Area of the Shaded Strip $= \frac{dE}{dT} \cdot dT = dE$

$\therefore \int_{T_1}^{T_2} dE = E = \text{Sum of the areas of all the strips between } NT_1 \text{ \& } QT_2$
 $= \text{Area } NT_1T_2Q.$

Thus the Seebeck e.m.f developed in the thermocouple for temp. difference $(T_2 - T_1)$ between the two junctions can be calculated from the area NT_1T_2Q from the graph.

Area of the rectangle $OT_1NM = T_1 \times \left(\frac{dE}{dT}\right)_{T_1} = \pi_1 \text{ --- (14)}$

Area of the rectangle $OT_2QP = T_2 \times \left(\frac{dE}{dT}\right)_{T_2} = \pi_2 \text{ --- (15)}$

Thus the Peltier coefficient of junctions A & B can be calculated from the area of the rectangles OT_1NM and OT_2QP from the graph.



From the power diagram:-

$$\text{area } OPQT_2 - \text{area } OMNT_1 = \pi_2 - \pi_1 = \text{area } MNT_1T_2 \& P$$

$$\therefore \text{area } MNT_1T_2 \& P - \text{area } NT_1T_2 \& = \text{area } MN \& PPM$$

$$\text{or } (\pi_2 - \pi_1) - E = \text{area } MN \& PPM.$$

$$\text{from eq}^n (5) \quad (\pi_2 - \pi_1) - E = - \int_{T_1}^{T_2} (\sigma_x - \sigma_y) dT$$

$$- \int_{T_1}^{T_2} (\sigma_x - \sigma_y) dT = \text{area } MN \& PPM \quad (6)$$

Let the element X of the thermocouple be taken as lead for which Thomson coefficient is zero. i.e. $\sigma_x = 0$

$$\int_{T_1}^{T_2} \sigma_y dT = \text{area } MN \& PPM.$$

[In this case the thermo-electric diagrams are for the thermocouple lead and the element Y]

$$\sigma_y (T_2 - T_1) = \text{Area } MN \& PPM \quad (17)$$

Thomson coefficient for the element Y can be calculated from (17).

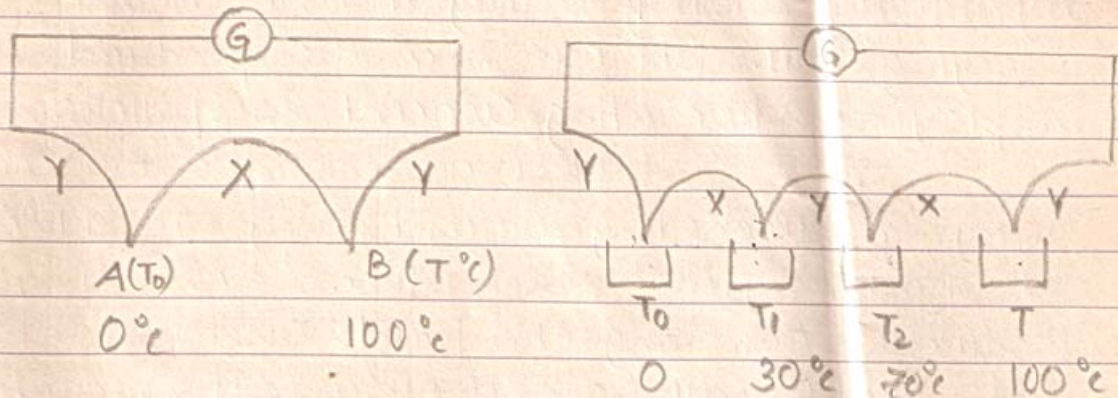
Similarly choosing a thermocouple made up of lead and the element X, we can draw the thermoelectric diagram & calculate σ_x .



Thomson Effect

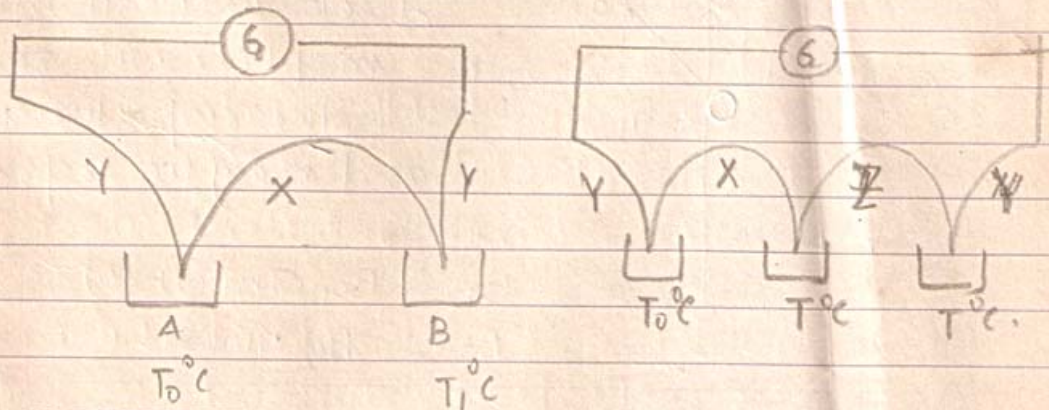
Laws of Thermoelectric circuit:

(1) Law of Successive temp:



$$E_{T_0}^T = E_{T_0}^{T_1} + E_{T_1}^{T_2} + E_{T_2}^T$$

(2) Law of Intermediate metals:



Introduction of a third metal does not effect the Seebeck emf in the circuit provide initial & final temp. of the junctions are same.